EMPIRICAL (SIMPLEST) FORMULA

1. Calculate the empirical formula of each of the compounds whose composition is given:

   a) 85.7% carbon; 14.3% hydrogen
   b) 52.9% aluminum; 47.1% oxygen
   c) 62.6 % lead; 8.4 % nitrogen; 29% oxygen

2. It was found by analysis that 8.20 g of a certain compound contains 3.45 g of sodium, 1.55 g of phosphorus and the remainder was oxygen. Calculate the simplest formula of the compound.

3. A sample of an organic compound called ninhydrin was found to be composed of 21.23 g of carbon, 1.18 g of hydrogen and 12.59 g of oxygen. Determine the simplest formula of ninhydrin.

4. When 2.61 g of aluminum were burned in oxygen, the resulting oxide had a mass of 4.921 g. Determine the simplest formula of aluminum oxide.

MOLECULAR FORMULA DETERMINATION

1. Determine the molecular formula for each of the following:

NAME	EMPIRICAL (SIMPLEST) FORMULA		MOLECULAR MASS (u)
a) ethylene glycol	CH_3O	= 31 $C_xH_3O_2$	62 $C_2H_6O_2$
b) octane	C_4H_9		114 C_8H_{18}
c) carotene	C_5H_7		536 $C_{40}H_{56}$

2. Two substances, one a gas and the other a liquid have the same percent composition: 92.3% carbon and 7.7% hydrogen. The molecular mass of the gas is 26 u and that of the liquid is 78 u. What is the molecular formula of each compound? gas - C_2H_2
   liquid - C_6H_6

3. A compound contains 92.9% carbon and 7.17% hydrogen. It's molecular mass is 78 u. Determine its molecular formula.

   C_6H_6

4. Upon analysis, a compound is found to contain 33% silicon and 67% fluorine. It's molecular mass is 170 u. Determine its molecular formula.

   Si_2F_6